Algorithmic Pseudocode Sample 47

Source-backed Image2Struct algorithm sample

1 Algorithm

This sample contains algorithmic pseudocode extracted from a source-backed LaTeX benchmark dataset. It is wrapped in a minimal article document for pdfLaTeX validation.

Algorithm 2 Square system method

Input:

- Algebraic system of difference equations named Σ'
- Time measured data allowing prolongation of the system.
- For $\bar{\mu} = \mu_1, \ldots, \mu_n$ the finite set of parameters, the data R_{μ_i} of permissible intervals for each parameter value.

Output: Parameter values of Σ'.

procedure DETECT SOLVABILITY

  Define $\Sigma prolong$ as an indefinite time prolongation of Σ'.

  Redefine $\Sigma' := []$, an 'empty system' of no equations.

  Denote $e_1, \ldots$ the equations of $\Sigma prolong$.

  Define $J(t) = \left[\dfrac{\partial e_i}{\partial \mu_j}\right]$ where $e_i \in \Sigma', i = 1, \ldots, r$, or 0 if Σ' is empty.

  $i := 1$

  for $rank(J(t)) := s < n$ i.e. is not of full rank, **do**

    procedure MOVE EQUATION

      $\Sigma' := \Sigma' \cup \{e_i\}$

      $\Sigma prolong := \Sigma prolong \backslash \{e_i\}$

      Compute $rank(J(t))$ (note Σ' has updated)

      if $rank(J(t)) < s + 1$ **then** $\Sigma' := \Sigma' \backslash \{e_i\}$

      end if

      $i = i + 1$

      if $rank(J(t) < n$ **then** repeat procedure Move Equation.

      end if

    end procedure

  end for

  procedure BLACKBOX SOLVER AND FILTER Note Detect Solvability runs until $J(t)$ has full rank and outputs a polynomial system Σ' that has solutions. Run any algebraic solver.

    Filter solutions by intersecting solution set with R_{μ_i}.

  end procedure

end procedure